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Cancer Incidence Trends Among Native Hawaiians and Other Pacific Islanders in the United States, 1990–2008

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Background Lack of annual population estimates for disaggregated Native Hawaiian and Other Pacific Islander (NHOPI) populations limits the ability to examine cancer incidence rates and trends to understand the cancer burdens among NHOPIs.

Methods Utilizing 1990 and 2000 population census data, we estimated the annual populations by age and sex for Native Hawaiians, Samoans, and Guamanians/Chamorros for 1990–2008 in regions covered by 13 of the National Cancer Institute's SEER registries. Cancer diagnoses during 1990–2008 from these registries were used to calculate the age-adjusted (2000 US Standard) incidence rates by sex, calendar year/period, and cancer type for each population. The annual percentage change (APC) in incidence rates was estimated with the 95% confidence intervals (95% CIs) calculated for both the rate and APC estimates.

Results Statistically significant declining trends were found in Native Hawaiians, in men for lung and stomach cancers (APC = –2.3%; 95% CI = –3.3 to –1.3; and APC = –3.8%; 95% CI = –6.0 to –1.6, respectively), and in women for breast cancer (APC = –4.1%; 95% CI = –5.7 to –2.5) since 1998 and lung cancer (APC = –6.4%; 95% CI = –10.7 to –1.8) since 2001. Rising incidence trends were experienced by Samoans, especially by Samoan women for breast (APC = 2.7%; 95% CI = 0.9 to 4.5) and uterus (APC = 7.3%; 95% CI = 6.2 to 8.4) cancers. With limited data, Guamanians/Chamorros demonstrated lower, but increasing, incidence rates than other NHOPIs.

Conclusions Population-based cancer incidence rates for disaggregated NHOPI populations help identify disparities in cancer burden and provide valuable information to improve cancer control efforts among NHOPIs.

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While substantial strides have been made in understanding cancer incidence patterns and trends among Asian American subgroups (1–10), much less is known about those among Native Hawaiian and Other Pacific Islander (NHOPI) populations in the United States (11–13). Despite the fact that population-based cancer registries in the United States routinely collect detailed NHOPI subgroup information, lack of information on populations at risk by ethnic subgroup has resulted in severe gaps in the knowledge about cancer burdens in these populations.

The NHOPI population is a small but fast-growing race group in the United States, representing 0.4% of the country's total population (14). The NHOPI population is an ethnically and culturally diverse group, including people with ancestry in any of the original peoples of Hawaii, Guam, Samoa, or other Pacific Islands. There were more NHOPIs of multirace (0.7 million) than there were NHOPIs of single race (0.5 million), according to the 2010 Census (14). Compared with the non-Hispanic (NH) white population, NHOPIs are younger, less educated, more likely to live in poverty, more likely to live in extended families, and less likely to be employed in management, professional, and high-income

occupations (15). There are marked demographic and socioeconomic differences among the NHOPI subgroups. For example, the median age was 31.8 years for Native Hawaiians, 24.4 for Samoans, and 29.3 for Guamanians; the poverty rate was 15.6% for Native Hawaiians, 20.2% for Samoans, and 13.7% for Guamanians (16).

To understand the cancer burden in specific minority populations and better target cancer research, control and prevention efforts, the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program (17) formed a working group to develop necessary annual population denominators and provide cancer incidence rates and trends for disaggregated Asian American and NHOPI groups. In this report, we present the results for the most common cancers among Native Hawaiians, Samoans, and Guamanians/Chamorros during 1990–2008.

Materials and Methods

Cancer Data

Cancer incidence data used in this study came from the population-based SEER registries. They represent new cancer cases diagnosed

between January 1, 1990 and December 31, 2008 reported to the 13 SEER registries as shown in Table 1.

Demographic (ie, age, sex, race/ethnicity) and clinical information on patients is routinely collected by the SEER registries from hospital medical records (18). Although the collection of multiple racial identities were instituted with cancer cases diagnosed in 2000 and onward, very few cases (less than 5%) have been reported with multiple races. Furthermore, the quality of the multiracial identity recorded in the registry database cannot be easily distinguished from simple recording errors. Therefore, only the primary race variable in the database was used in grouping the cancer cases by racial/ethnic categories. SEER coding rules require the minority race to be the primary when more than one race is reported, to ensure the representation of smaller populations in cancer registry data (18). The cancer types were classified according to the *International Classification of Diseases for Oncology, Third Edition* (ICD-O-3) (19), and regrouped into major categories for analysis according to the SEER site recode (20).

A total of 22 720 NHOPI cancer cases were reported to the 13 registries during 1990–2008. We excluded cancer cases with missing or unknown age (1 [0.00%]), missing or unknown sex (3 [0.01%]), unspecific NHOPI identification (1102 [4.85%]), unavailability of population counts across census years for some specific NHOPI subgroups including Tongans (2051 [9.03%]), and withholding of area population less than 100 by 2000 Census (64 [0.28%]). A total of 19 499 (85.8% of the total reported cases) remained for analysis, including 16 062 Native Hawaiians, 2760 Samoans, and 677 Guamanians/Chamorros. For comparison purposes, the 4099 552 non-Hispanic (NH) white cancer cases from the same registries diagnosed during the same time period were also included in the study.

Population Estimates

We used the Summary Tape File 2 (STF2) of the 1990 Census and Summary File 2 (SF2) of the 2000 Census to obtain the detailed population counts for specific NHOPI subgroups of Native Hawaiians, Samoans, and Guamanians/Chamorros. Because the 2000 Census allowed multiple identifications of race for an individual (21), a bridging method was needed to adjust the 2000 multiracial classification scheme into mutually exclusive single race categories to be compatible with the 1990 data. We bridged the 2000 Census results from its multirace scheme into a single race scheme for all SEER regions by taking the simple average between the minimum population count (ie, people who reported one race) and the maximum count (ie, people who reported more than one race) for a given racial/ethnic group. The age-sex structures of each of the NHOPI subgroups (in percentages of the total NHOPI population) from the 1990 Census and the bridged 2000 Census (April) data were applied to the midyear (July) 1990 and 2000 Census Bureau estimates for total NHOPIs to obtain the midyear population estimates for 1990 and 2000, respectively, for the specific NHOPI subgroups. The 1991–1999 age-sex subgroup-specific population estimates were developed using linear interpolations between the 1990 and 2000 midyear estimates, and the 2001–2008 estimates were extrapolated using this linear model.

We adopted the Hawaii Tumor Registry (HTR) 1990–2005 population estimates and extrapolated linearly for 2006–2008 for the NHOPI subgroups in Hawaii, as a modification to the above-described population estimation method. Before the 2000 Census that adopted a multiracial classification scheme, the racial/ethnic classification from the census was too crude to accurately categorize Hawaii's multiracial populations. To correct for undercounting of the

Table 1. Distributions of cancer cases and population estimates by racial/ethnic groups and Surveillance, Epidemiology, and End Results (SEER) regions, 1990–2008

SEER region	Non-Hispanic white		Native Hawaiian		Samoan		Guamanian/Chamorro	
	Case	Population	Case	Population	Case	Population	Case	Population
All	4 099 552	721 057 524	16 062	5 729 112	2760	1 381 197	677	728 731
	100.0%	100.0%	99.9%	100.0%	99.6%	100.0%	99.0%	100.0%
California*	1 821 989	312 423 738	1526	763 872	1383	816 372	469	522 268
	44.4%	43.3%	9.5%	13.3%	50.1%	59.1%	69.3%	71.7%
Connecticut	312 154	50 909 380	19	10 444	†	6076	†	6418
	7.6%	7.1%	0.1%	0.2%	†	0.4%	†	0.9%
Hawaii	29 612	5 227 124	14 007	4 730 170	1050	285 883	72	54 373
	0.7%	0.7%	87.2%	82.6%	38.0%	20.7%	10.6%	7.5%
Iowa	286 220	51 542 709	21	8854	†	3788	†	4171
	7.0%	7.1%	0.1%	0.2%	†	0.3%	†	0.6%
New Jersey	692 332	106 637 859	213	19 671	15	12 685	12	17 772
	16.9%	14.8%	1.3%	0.3%	0.5%	0.9%	1.8%	2.4%
New Mexico	89 670	15 636 347	28	15 975	†	6626	†	8026
	2.2%	2.2%	0.2%	0.3%	†	0.5%	†	1.1%
Utah	116 870	36 332 597	19	45 119	60	98 273	†	5063
	2.9%	5.0%	0.1%	0.8%	2.2%	7.1%	†	0.7%
Atlanta	126 726	28 644 708	†	3011	†	†	†	†
(metropolitan)	3.1%	4.0%	†	0.1%	†	†	†	†
Detroit	302 508	52 845 723	†	5021	†	2762	†	†
(metropolitan)	7.4%	7.3%	†	0.1%	†	0.2%	†	†
Seattle/Puget	321 471	60 857 339	220	126 974	242	148 732	117	110 641
Sound	7.8%	8.4%	1.4%	2.2%	8.8%	10.8%	17.3%	15.2%

* Including Los Angeles, San Francisco/Oakland and San Jose/Monterey, and the Greater California registries.

† Data withheld due to case count <10.

Native Hawaiian population in the 1970, 1980, and 1990 censuses, the HTR developed its own population estimates based on survey data collected by the Hawaii Department of Health for the detailed racial/ethnic populations in Hawaii for 1975–2005. The HTR population estimates have been routinely used by the HTR and Hawaii Department of Health for generating all cancer statistics by detailed race/ethnicity for the state of Hawaii (22–23). The methodologies used to create the HTR population estimates were derived by apportioning the census sex-age counts by the race distributions for each sex-age group as determined by the annual Health Habits Survey, after adding the census-determined group quarters data. Given that the ethnic distributions from the Health Habits Survey varied from year to year, especially for small population groups, a weighted running average method by sex, age, and race/ethnicity was used to get a smoother center value. Further unpublished technical notes of the HTR estimates are available upon request.

Because of the Census Bureau's data withholding rules, we were unable to obtain necessary population denominators for certain NHOPI subgroups in some of the SEER reporting areas where the total population of a group was less than 100 in the 2000 Census SF2 (21). In these situations, an entire registry or a county of an registry was excluded from the analysis for that specific NHOPI subgroup.

Statistical Methods

The average annual age-adjusted (2000 US Standard) incidence rates per 100 000 population and 95% confidence intervals (CIs) were generated using SEER*Stat software (<http://seer.cancer.gov/seerstat/>) for the top five most common cancers for NH whites, Native Hawaiians, and Samoans for three time periods (1990–1995, 1996–2001, and 2002–2008). For Guamanians/Chamorros, incidence rates are shown for two time periods (1990–1999 and 2000–2008). Diagnosis years were combined as necessary to ensure at least 10 cancer cases per time period for the stability of the rate estimates (24). Joinpoint regression models (<http://surveillance.cancer.gov/joinpoint/>) were used to analyze the trends of annual or average annual incidence rates for NH whites, Native Hawaiians, and Samoans, respectively, over the entire study period (1990–2008), and the annual percentage change (APC) and 95% confidence interval was estimated for each trend line or segment. These confidence intervals provide more information about the direction and strength of the estimated effect, and statistical significance can be assumed if the confidence interval does not contain zero. A maximum of three joinpoints were allowed based on single-year data, two joinpoints based on two-year grouping, and no joinpoints could be estimated for trends based on five-year grouping due to limited number of data points. When no joinpoints were detected over time for a given group, the APC for the entire study period was provided. Plots of rate estimates and trend lines produced by the joinpoint analysis are shown in semi-logarithmic scale and 3:1 y:x-axis aspect ratio for comparison of changes by cancer type among groups (25–26).

Results

There are obvious geographic clusterings of NHOPI populations among SEER registries. Whereas 82.6% of Native Hawaiians live in Hawaii, the majority of Samoans and Guamanians/Chamorros

live outside of Hawaii. More Samoans and Guamanians/Chamorros live in California than in Hawaii. Most of the NHOPI cancer cases came from the SEER registries in Hawaii, California, and Seattle/Puget Sound (Washington).

During 1990–2008, similar to NH whites, prostate (in men), breast (in women), lung, and colorectal (CRC) cancers were the top three most common in almost all NHOPI subgroups examined in both sexes, except CRC that was ranked fourth in Samoan women (Tables 2 and 3). However, in contrast to their lower rankings in NH whites, stomach (fourth in Native Hawaiian and Samoan men and fifth in Samoan men), liver (fifth in Samoan men), and pancreatic (fifth in Native Hawaiian women) cancers were among the leading cancers in the NHOPI populations. The incidence rates and trends varied markedly among NHOPIs by cancer type and population subgroup.

Native Hawaiians

Among Native Hawaiian men, incidence trends for the top five cancers (prostate, lung, CRC, stomach, and non-Hodgkin lymphoma) were declining, with statistically significant decreases in lung and stomach cancers (–2.3% [95% CI = –3.3 to –1.3 and –3.8%, 95% CI = –6.0 to –1.6, respectively) (Figure 1). Their risks for lung and stomach cancers were consistently higher than those of the NH white men over time, but the differences narrowed with the faster pace of decline in Native Hawaiian men. After slightly rising during the early years, their incidence trends for all cancers combined started to decline with a statistically significant APC of –2.7 (95% CI = –4.3 to –1.1) during 2000–2008 (Figure 1 and Table 2). Declining incidence trends were also observed among Native Hawaiian women for all of the five most common cancers (breast, lung, CRC, uterus, and pancreas) and all sites combined (Figure 2 and Table 3). The decreasing APCs since 1998 for all sites (Figure 2; –3.3%; 95% CI = –4.3 to –2.4) and breast (Figure 2; –4.1%; 95% CI = –5.7 to –2.5) cancers and since 2001 for lung (Figure 2; –6.4%; 95% CI = –10.7 to –1.8) reached statistical significance. They also experienced consistently higher incidence rates for uterus and pancreatic cancers over the years than their NH white counterparts (Figure 2 and Table 3).

Samoans

In contrast to the decreasing trends in prostate and CRC cancers in NH white and Hawaiian men, both were on the rise for Samoan men (Figure 1 and Table 2). Samoan men consistently had higher incidence rates for lung and stomach cancers than NH white and Native Hawaiian men. Statistically significant APCs of incidence rates were found in Samoan women for breast (2.7%; 95% CI = 0.9 to 4.5) (Figure 2) and uterine (7.3%; 95% CI = 6.2 to 8.4) cancers. Despite the substantial decrease in incidence rates in both men and women, Samoans still suffered from the disproportionately higher stomach cancer risk than NH whites and Native Hawaiians (Tables 2 and 3, Figures 1 and 2).

Guamanians/Chamorros

Limited by small case counts, rates for Guamanians/Chamorros could only be calculated for two broadly grouped time periods (1990–1999 and 2000–2008) as included in Tables 2 and 3 and for the top three cancer sites in men (prostate, lung, CRC) and women

Table 2. Cancer case counts and age-adjusted incidence rates with 95% confidence interval (CI) of the most common cancers in Native Hawaiian and Other Pacific Islander (NHOPI) subgroups and non-Hispanic whites by time period, males*

Race/Ethnicity	Rank	1990–1995			1996–2001			2002–2008		
		Count	Rate	95% CI	Count	Rate	95% CI	Count	Rate	95% CI
Non-Hispanic white										
All Sites		655 690	620.9	(619.3 to 622.4)	665 295	590.0	(588.1 to 590.9)	800 107	565.2	(564.0 to 566.5)
Prostate	1	202 261	191.7	(190.8 to 192.5)	195 821	172.2	(171.4 to 173.0)	229 665	158.6	(158.0 to 159.3)
Lung	2	101 781	95.7	(95.1 to 96.3)	96 883	85.8	(85.3 to 86.4)	105 743	75.5	(75.1 to 76.0)
CRC	3	73 143	71.3	(70.7 to 71.8)	73 894	66.5	(66.0 to 67.0)	78 328	55.8	(55.4 to 56.2)
NHL	6	26 220	24.4	(24.1 to 24.8)	28 554	25.2	(24.9 to 25.5)	36 092	25.7	(25.5 to 26.0)
Stomach	11	12 032	11.8	(11.6 to 12.0)	11 604	10.5	(10.3 to 10.7)	12 543	9.0	(8.8 to 9.1)
Liver	16	5 283	5.0	(4.9 to 5.1)	7 287	6.4	(6.3 to 6.6)	11 832	8.0	(7.9 to 8.2)
Native Hawaiian										
All sites		1953	484.8	(461.5 to 508.8)	2399	497.0	(475.5 to 518.4)	3126	432.9	(417.3 to 448.9)
Prostate	1	395	114.6	(102.7 to 127.4)	510	119.3	(108.6 to 130.7)	685	104.2	(96.4 to 112.5)
Lung	2	417	103.4	(93.1 to 114.4)	460	98.6	(89.3 to 108.5)	545	78.7	(72.0 to 85.8)
CRC	3	241	62.1	(53.8 to 71.2)	302	63.4	(56.1 to 71.4)	410	55.8	(50.4 to 61.7)
Stomach	4	92	24.2	(19.3 to 30.0)	94	20.2	(16.1 to 25.0)	96	14.2	(11.5 to 17.4)
NHL	5	79	18.6	(14.2 to 23.8)	97	17.8	(14.2 to 22.0)	132	17.4	(14.5 to 20.8)
Samoan										
All sites		317	576.8	(502.8 to 657.3)	384	475.5	(421.6 to 533.8)	630	542.1	(493.9 to 593.2)
Prostate	1	64	163.6	(121.9 to 213.0)	83	120.9	(93.9 to 152.5)	148	169.0	(140.4 to 201.0)
Lung	2	69	120.4	(90.8 to 155.9)	79	100.6	(77.6 to 127.9)	108	98.9	(79.0 to 121.8)
CRC	3	28	42.3	(26.2 to 64.0)	40	47.4	(31.3 to 68.2)	71	45.2	(34.1 to 58.8)
Stomach	4	28	71.3	(43.3 to 108.0)	29	44.1	(27.7 to 65.5)	35	25.5	(17.1 to 36.6)
Liver	5	18	29.7	(16.2, 49.1)	27	30.1	(18.8 to 45.4)	34	30.7	(19.9 to 44.7)
2000–2008										
Guamanian/ Chamorro†										
All sites		143	201.9	(164.1 to 245.0)	213	228.3	(194.7 to 265.6)			
Prostate	1	32	65.1	(41.9 to 95.0)	65	87.1	(65.1 to 113.3)			
Lung	2	27	37.9	(23.9 to 56.9)	34	37.7	(25.2 to 53.9)			
CRC	3	11	17.4	(7.6 to 33.1)	28	28.7	(17.8 to 43.5)			

* Rank based on overall rate from 1990 to 2008 for all areas combined. Incidence rates are age-adjusted to the 2000 US Standard population and presented for invasive cancers only. Abbreviations: Lung = lung and bronchus; Liver = liver and intrahepatic bile duct; CRC = colon and rectum; NHL = non-Hodgkin lymphoma.

† Data for fourth- and fifth-ranked cancers withheld due to cancer case count <10.

Table 3. Cancer case counts and age-adjusted incidence rates with 95% confidence interval (CI) of the most common cancers in Native Hawaiian and Other Pacific Islander (NHOPI) subgroups and non-Hispanic whites by time period, females*

Race/Ethnicity	Rank	1990–1995			1996–2001			2002–2008		
		Count	Rate	95% CI	Count	Rate	95% CI	Count	Rate	95% CI
Non-Hispanic white										
All sites		589 274	437.0	(435.9 to 438.1)	637 140	453.0	(451.8 to 454.1)	752 046	441.0	(440.0 to 442.0)
Breast	1	183 463	140.8	(140.1 to 141.4)	203 032	148.4	(147.7 to 149.0)	228 639	136.6	(136.0 to 137.2)
Lung	2	76 529	55.4	(55.0 to 55.8)	83 761	57.8	(57.4 to 58.2)	100 847	56.9	(56.6 to 57.3)
CRC	3	71 404	49.5	(49.2 to 49.9)	72 581	47.7	(47.4 to 48.1)	76 608	41.8	(41.5 to 42.1)
Uterus	4	35 755	26.8	(26.5 to 27.0)	37 663	27.2	(26.9 to 27.5)	44 557	26.0	(25.8 to 26.3)
Pancreas	11	14 399	10.0	(9.8 to 10.1)	14 965	9.8	(9.6 to 10.0)	19 291	10.4	(10.3 to 10.6)
Stomach	16	7 284	5.0	(4.9 to 5.1)	6 872	4.5	(4.4 to 4.6)	7 224	3.9	(3.8 to 4.0)
Native Hawaiian										
All sites		2 175	459.1	(439.0 to 479.8)	2 706	478.1	(459.7 to 497.1)	3 703	394.2	(381.6 to 407.2)
Breast	1	721	148.3	(137.2 to 159.9)	1 010	173.1	(162.4 to 184.4)	1 284	136.8	(129.4 to 144.5)
Lung	2	267	61.2	(53.8 to 69.3)	333	64.5	(57.6 to 71.9)	466	51.6	(47.0 to 56.5)
CRC	3	185	43.3	(37.0 to 50.2)	236	44.9	(39.2 to 51.1)	289	31.3	(27.8 to 35.1)
Uterus	4	179	35.5	(30.4 to 41.3)	217	35.6	(30.9 to 40.8)	340	35.7	(32.0 to 39.7)
Pancreas	5	59	14.3	(10.7 to 18.6)	83	16.4	(13.0 to 20.4)	124	13.5	(11.2 to 16.1)
Samoan										
All sites		299	389.8	(340.7 to 443.4)	415	391.2	(350.6 to 434.9)	715	454.6	(418.9 to 492.3)
Breast	1	72	86.4	(65.9 to 110.9)	103	91.3	(73.5 to 112.0)	182	114.8	(97.6 to 134.0)
Uterus	2	34	35.6	(23.5 to 51.5)	60	47.3	(35.3 to 62.1)	144	85.0	(70.8 to 101.2)
Lung	3	29	48.4	(31.1 to 70.8)	45	52.9	(37.6 to 71.6)	60	43.6	(32.6 to 56.7)
CRC	4	13	14.8	(6.7 to 27.8)	19	20.8	(11.9 to 33.2)	60	40.7	(30.1 to 53.6)
Stomach	5	24	41.3	(25.0 to 63.0)	21	24.0	(14.0 to 37.8)	24	18.7	(11.5 to 28.2)
2000–2008										
Guamanian/ Chamorro†										
All sites		130	153.2	(124.3 to 186.3)	191	171.6	(146.1 to 200.1)			
Breast	1	36	35.8	(24.0 to 51.1)	57	46.8	(34.9 to 61.5)			
Lung	2	21	26.6	(15.7 to 41.7)	26	26.1	(16.6 to 38.6)			
CRC	3	10	19.8	(8.5 to 37.0)	13	11.5	(5.9 to 20.1)			

* Rank based on overall rate from 1990 to 2008 for all areas combined. Incidence rates are age-adjusted to the 2000 US standard population and presented for invasive cancers only. Abbreviations: Lung = lung and bronchus; CRC = colon and rectum; Uterus = corpus uteri and uterus; not otherwise specified.

† Data for fourth- and fifth-ranked cancers withheld due to cancer case count <10.

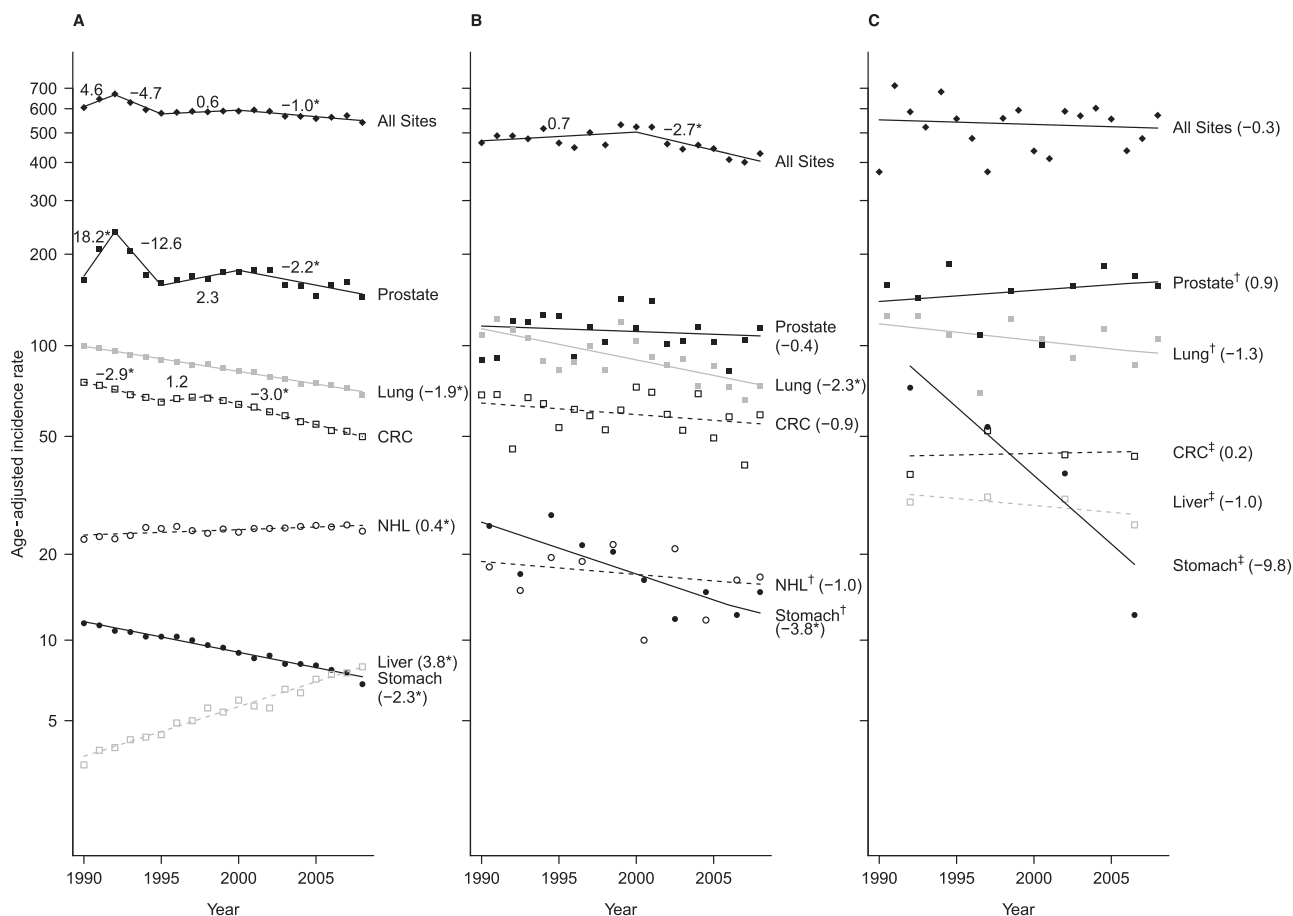


Figure 1. Trends of age-adjusted incidence rates and annual percentage change (APC) by racial/ethnic group for the five most common cancers among males, 1990–2008. **A)** Trends for non-Hispanic whites are shown. The 95% confidence intervals are as follows: All Sites segment 1: –0.8 to 10.2, segment 2: –9.4 to 0.3, segment 3: –1.0 to 2.2, segment 4: –1.5 to –0.4; Prostate segment 1: 1.1 to 38.1, segment 2: –25.1 to 2.0, segment 3: –2.8 to 7.7, segment 4: –3.8 to –0.6; Lung: –2.0 to –1.8; CRC segment 1: –3.4 to –2.4, segment 2: –1.0 to 3.5, segment 3: –3.2 to –2.8; NHL: 0.2 to 0.6; Liver: 3.5 to 4.2; Stomach: –2.5 to –2.1. **B)** Trends for Native Hawaiians are shown. The 95% confidence intervals are as follows: All Sites segment 1: –0.7 to

2.1, segment 2: –4.3 to –1.1; Prostate: –1.7 to 0.9; Lung: –3.3 to –1.3; CRC: –2.2 to 0.5; NHL: –4.0 to 2.2; Stomach: –6.0 to –1.6. **C)** Trends for Samoans are shown. The 95% confidence intervals are as follows: All Sites: –1.9 to 1.2; Prostate: –2.0 to 3.8; Lung: –3.4 to 0.9; CRC: –5.7 to 6.6; Liver: –4.6 to 2.6; Stomach: –2.1 to 3.1. *Statistically significantly different APCs. †Joinpoint and observed rates are based on 2-year groups (1990–1991, 1992–1993, ..., 2004–2005, 2006–2007, 2008). ‡Joinpoint and observed rates are based on five-year groups (1990–1994, 1995–1999, 2000–2004, 2005–2008). CRC = colon and rectum, Liver = liver and intrahepatic bile duct, Lung = lung and bronchus, NHL = non-Hodgkin lymphoma.

(breast, lung, CRC), respectively. No joinpoint analysis could be done because of too few data points. Although these top ranked cancers were the same as in the other populations examined, Guamanians/Chamorros demonstrated much lower incidence rates than those of Native Hawaiians and Samoans in both men and women. However, their cancer risks appeared to increase over time, except for lung cancer (Tables 2 and 3).

Geographic Differences

As shown in Table 4, Native Hawaiian men residing in Hawaii displayed higher incidence rates for lung cancer but lower incidence rates for prostate and CRC cancers than did Native Hawaiian men living on the mainland. Rates for all cancers combined declined more dramatically for those on the mainland than in Hawaii. Native Hawaiian women in Hawaii appeared to have higher risk for almost all cancers examined (especially for breast, lung, uterine, and all cancers combined) than Native Hawaiian women living on the mainland.

Discussion

With a total of nearly 20 000 Native Hawaiian, Samoan, and Guamanian/Chamorro cancer cases from 13 SEER registries during the period 1990–2008, our study highlights several important findings: 1) There are clear differences in cancer incidence rates among NHOPI subgroups and between NHOPIs and NH whites by cancer type; 2) the cancer-specific incidence trends vary by degree and direction among NHOPI groups; 3) rising cancer trends in Samoans and Guamanians/Chamorros are of particular concern; and 4) cancer incidence rates and trends differed between Native Hawaiians in Hawaii and those on the mainland.

NHOPIs share the same burden of common cancers such as prostate, breast, lung, and CRC as NH whites, but they also bear higher risks from cancers including stomach, liver, and uterus. These patterns are consistent with previous findings (23,27–28).

We observed that female breast cancer incidence trends increased during the 1990s, then decreased afterward among Native Hawaiians but continued to rise among Samoans and Guamanians/

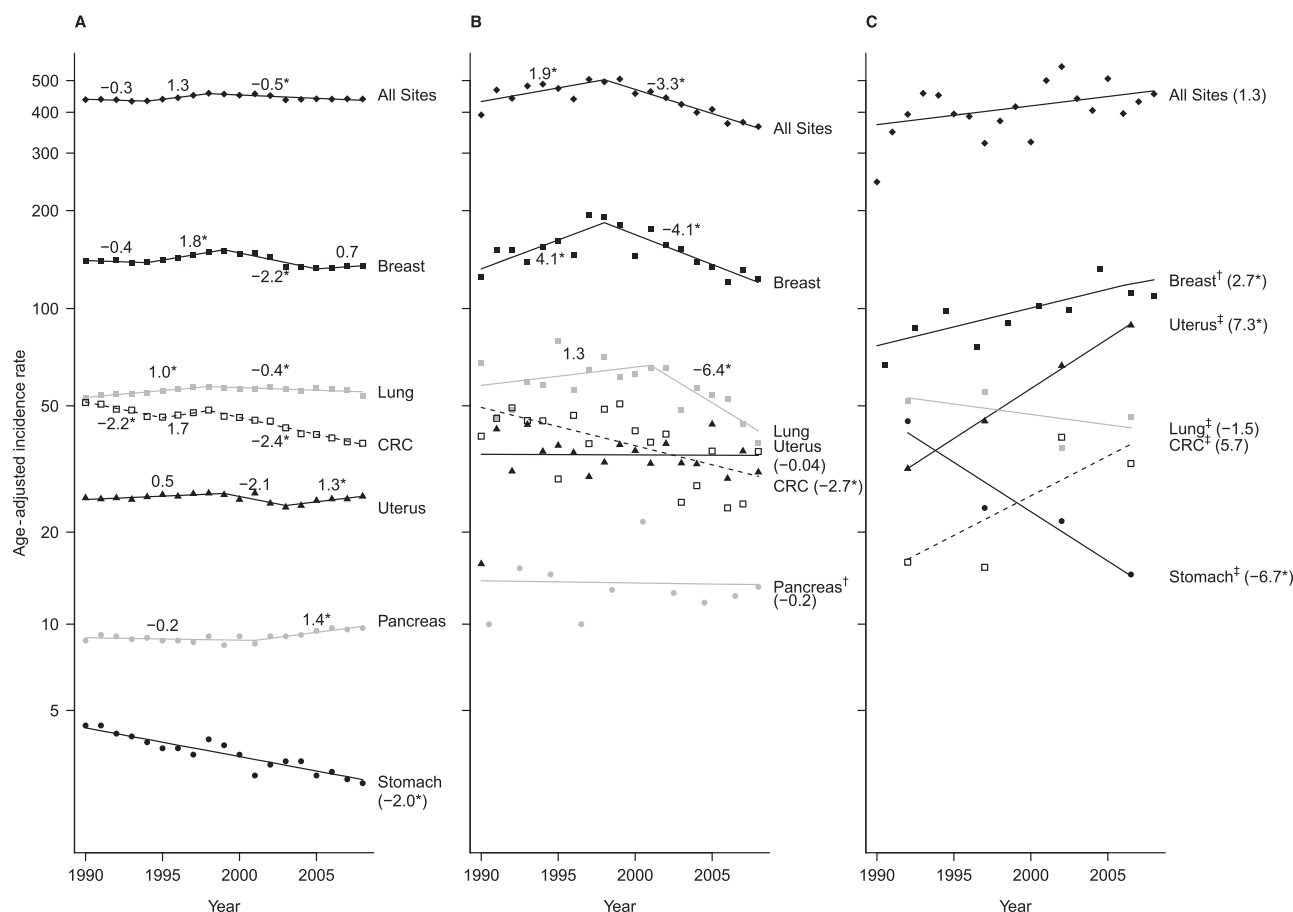


Figure 2. Trends of age-adjusted incidence rates and annual percent-age change (APC) by racial/ethnic group for the five most common cancers among females, 1990–2008. **A)** Trends for non-Hispanic whites are shown. The 95% confidence intervals are as follows: All Sites segment 1: –1.2 to 0.6, segment 2: –0.1 to 2.8, segment 3: –0.7 to –0.2; Breast segment 1: –2.0 to 1.3, segment 2: 0.3 to 3.4, segment 3: –3.3 to –1.1, segment 4: –1.9 to 3.4; Lung segment 1: 0.6 to 1.3, segment 2: –0.6 to –0.1; CRC segment 1: –3.0 to –1.4, segment 2: –1.7 to 5.1, segment 3: –2.6 to –2.1; Uterus segment 1: –0.1 to 1.1, segment 2: –4.6 to 0.5, segment 3: 0.1 to 2.5; Pancreas segment 1: –0.5 to 0.2, segment 2: 0.7 to 2.1; Stomach: –2.4 to –1.6. **B)** Trends for Native Hawaiians are shown. The 95% confidence intervals are as follows: All Sites segment 1: 0.3 to 3.6,

segment 2: –4.3 to –2.4; Breast segment 1: 1.2 to 7.2, segment 2: –5.7 to –2.5; Lung segment 1: –1.6 to 4.3, segment 2: –10.7 to –1.8; Uterus: –1.5 to 1.5; CRC: –4.3 to –1.0; Pancreas: –3.4 to 3.2. **C)** Trends for Samoans are shown. The 95% confidence intervals are as follows: All Sites: –0.1 to 2.8; Breast: 0.9 to 4.5; Uterus: 6.2 to 8.4; Lung: –8.1 to 5.7; CRC: –9.1 to 22.9; Stomach: –12.5 to –0.5. *Statistically significantly different APCs. †Joinpoint and observed rates are based on two-year groups (1990–1991, 1992–1993, ..., 2004–2005, 2006–2007, 2008). ‡Joinpoint and observed rates are based on five-year groups (1990–1994, 1995–1999, 2000–2004, 2005–2008). CRC = colon and rectum, Liver = liver and intrahepatic bile duct, Lung = lung and bronchus, Uterus = corpus uteri and uterus not otherwise specified.

Chamorros. Similar increases in breast cancer incidence during 1991–1996 have been reported by the Multiethnic Cohort (MEC) study (29). A subsequent decline in breast cancer incidence rates among Native Hawaiians was reported in the population-based cancer surveillance system in Hawaii (23), as well as within the MEC database (personal communication with LR Wilkens, February 2, 2013). Because the MEC is a cohort, incidence rates can be computed within the cohort; no external population estimates are required. Thus, the comparability between our findings and the MEC trends gave us confidence in our findings. Such changes in breast cancer trends may be attributable to the change in prevalence of breast cancer risk factors among Native Hawaiian women, who in recent years displayed reduced smoking and drinking prevalence and higher utilization of physical exams and screening (30–31). Challenges with regard to knowledge, attitude, and recently improved access to breast cancer screening among the Samoan and Guamanian/Chamorro communities may underlie the rising breast cancer trends in these populations (32–35).

The declining incidence trends in lung and stomach cancers among Native Hawaiians and Samoans parallel national (36) and local (23) trends that were accompanied by reductions in smoking prevalence (37). Rising trends in prostate, breast, uterine, and CRC cancers among Samoans and Guamanians/Chamorros warrant focused attention. These cancers are frequently associated with obesity, especially later in life (38–39). Obesity is highly prevalent in NHOPI populations (39–41).

When examined separately by geographic location, we found that Native Hawaiians living in Hawaii experienced higher risks for lung, breast, and uterus cancers, but lower risks for CRC and prostate cancers, as compared to the Native Hawaiians living on the US mainland. These geographic differences in cancer rates for Native Hawaiians have not been previously reported. In a different type of comparison, an earlier study compared the relative risk for several cancers between Samoans and Native Hawaiians living in Los Angeles vs in Hawaii and found different patterns by geography (42). There may well be a web of demographic,

Table 4. Cancer case counts and age-adjusted incidence rates with 95% confidence interval (CI) of the top five cancers among Native Hawaiians by sex, time period, and geographic location*

		1990–1995				1996–2001			2002–2008		
Sex	Rank		Count	Rate	95% CI	Count	Rate	95% CI	Count	Rate	95% CI
Males											
All sites		Hawaii	1690	489.3	(464.4 to 515)	2077	509.0	(486.1 to 532.9)	2720	448.0	(430.7 to 465.8)
		Mainland	263	513.5	(439.3 to 595.1)	322	453.0	(398.2 to 512.9)	406	393.0	(351.9 to 437.2)
Prostate	1	Hawaii	324	108.3	(96 to 121.5)	420	117.0	(105.5 to 129.2)	566	102.7	(94.2 to 111.6)
		Mainland	71	165.0	(123.5 to 214.2)	90	130.7	(102.3 to 163.8)	119	117.1	(95.7 to 141.5)
Lung	2	Hawaii	381	110.1	(98.8 to 122.3)	411	104.0	(93.8 to 114.9)	497	85.9	(78.3 to 93.9)
		Mainland	36	79.9	(52 to 115.7)	49	72.2	(51 to 98.2)	48	49.2	(34.7 to 67)
CRC	3	Hawaii	197	57.8	(49.5 to 67)	260	64.7	(56.7 to 73.4)	345	55.4	(49.5 to 61.7)
		Mainland	44	109.0	(73.8 to 152.9)	42	62.8	(43.1 to 87.4)	65	69.9	(52.2 to 90.9)
Stomach	4	Hawaii	84	26.2	(20.7 to 32.7)	81	20.1	(15.7 to 25.2)	84	14.8	(11.7 to 18.3)
		Mainland	†	†	†	13	24.5	(12.1 to 42.6)	12	13.7	(6.5 to 24.4)
NHL	5	Hawaii	67	18.4	(13.8 to 23.9)	80	18.0	(14.1 to 22.6)	111	17.5	(14.3 to 21.2)
		Mainland	12	20.2	(7.8 to 40.8)	17	15.3	(8.7 to 25.5)	21	20.0	(11.5 to 31.7)
Females											
All sites		Hawaii	1916	478.8	(456.7 to 501.7)	2363	499.9	(479.5 to 521)	3241	408.8	(394.7 to 423.1)
		Mainland	259	355.7	(308.2 to 407.9)	343	382.2	(339.4 to 428.5)	462	348.7	(315.6 to 384.1)
Breast	1	Hawaii	626	153.6	(141.5 to 166.5)	882	182.3	(170.3 to 194.9)	1125	142.9	(134.6 to 151.5)
		Mainland	95	121.1	(95.9 to 150.7)	128	133.7	(109.7 to 161)	159	116.2	(97.9 to 136.7)
Lung	2	Hawaii	242	64.7	(56.6 to 73.6)	289	66.5	(58.9 to 74.8)	423	54.6	(49.5 to 60.1)
		Mainland	25	39.7	(24.1 to 60.9)	44	55.0	(39 to 74.8)	43	36.3	(25.8 to 49.4)
CRC	3	Hawaii	162	44.7	(37.8 to 52.3)	199	45.2	(39 to 52)	246	31.3	(27.5 to 35.4)
		Mainland	23	34.8	(20.5 to 54.5)	37	43.6	(30 to 60.9)	43	39.6	(28.1 to 53.8)
Uterus	4	Hawaii	160	38.0	(32.2 to 44.5)	183	36.2	(31 to 41.9)	298	37.7	(33.5 to 42.2)
		Mainland	19	24.0	(14.1 to 38.3)	34	34.4	(23.1 to 49)	42	27.0	(19.2 to 36.9)
Pancreas	5	Hawaii	52	14.7	(10.8 to 19.4)	72	16.9	(13.1 to 21.3)	110	13.9	(11.4 to 16.7)
		Mainland	†	†	†	11	14.2	(6.8 to 25.5)	14	14.0	(7.5 to 23.3)

* Rank based on overall rate from 1990 to 2008 for all areas combined. Incidence rates are age-adjusted to the 2000 US standard population and presented for invasive cancers only. Abbreviations: Lung = lung and bronchus; CRC = colon and rectum; NHL = non-Hodgkin lymphoma; Uterus = corpus uteri and uterus, not otherwise specified.

† Data withheld due to cancer case count <10.

social, behavioral, cultural, and disease factors at work that may explain the mainland vs Hawaiian differences. Studies have shown that Native Hawaiian migrants experience similar cultural adjustment as immigrants (43–44). The migration and westernization experience may result in the same dietary changes and health consequences regardless of the diverse NHOPI cultural backgrounds (45–46). Alternatively, these patterns may reflect selective migration such that healthier individuals are more likely to migrate over less healthy individuals—the “healthy migrant effect” (47–48), or that more highly educated professionals with better health behaviors are more likely to migrate (49–50). Examination of cancer stage distribution by geographic location did not reveal notable differences (data available upon request), suggesting that the observed differences in incidence rates are likely not attributable to differences in cancer screening practices. Future research designed to understand the causes for this geographic differences will likely provide valuable insights for reducing cancer disparities among Native Hawaiians.

Our study is not without limitations. The calculation of cancer rates depends on accurate classification of cancer cases into the specific racial/ethnic subgroups and reliable estimates of annual population at risk. In our analysis, we used the SEER data, which represent the highest quality of population-based cancer registry records. However, previous studies have revealed misclassification issues in identifying NHOPI cancer cases in cancer registry data that may slightly affect the incidence rate estimates (51–54). In estimating

the population denominators, we also relied on a set of assumptions regarding the population growth based on the decennial census data. Other studies using different population estimates will yield incidence rates that are not directly comparable to our results. To test the effect of adopting the HTR population estimates methods, we employed a sensitivity analysis using population estimates that were free of the HTR adjustment for census undercounts. Using this alternative set of denominators, the observed trends continue to hold (data available upon request). This sensitivity analysis, along with the agreement with MEC data, indicates the robustness of the population estimates used in the study. Furthermore, the exclusion of the 1102 (4.85%) NHOPI cancer cases from our analysis due to lack of specific subgroup designation may have resulted in a slight underestimate of cancer rates for each of the NHOPI subgroups in the study. Different methods and assumptions for identifying Native Hawaiians in hospital records vs populations censuses, as well as between the HTR and mainland SEER registries, may have affected the rate estimates for this group. Therefore, the observed differences in cancer-specific risks between Native Hawaiians living in Hawaii and those living on the US mainland should be interpreted with caution. Furthermore, the relatively small numbers of population and cancer cases for NHOPI in general and for Samoans and Guamanians/Chamorros in particular may cause unstable rate estimates. Due to the small numbers, we could not compare risk difference by geographic location for Samoans and Guamanians/Chamorros as we did for Native Hawaiians, even though the

majority of these populations covered by SEER registries actually live on the mainland. Continued cancer surveillance and focused study of individual NHOPI subpopulations is not only necessary, but also important.

In summary, our study provides much-needed disaggregated information regarding the cancer burden and disparities within the NHOPI populations with novel observations that bear important policy implications. Continued effective and targeted cancer control efforts are needed in all NHOPI populations, especially the Samoan and Guamanian/Chamorro communities. We hope the findings will be useful to promote better cancer control programs that aim not only at screening, but also culturally sensitive education and outreach in the NHOPI communities and migrants.

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Notes

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